

Zeshan M. Hussain

Curriculum Vitae

Research Interests

My research studies learning and decision-making in partially-observed, intervention-driven environments, using oncology as a rigorous test bed. I develop methods spanning representation learning, causal inference, sequential decision-making, and physician-AI collaboration, with the goal of building adaptive AI systems that can model a patient's latent disease dynamics, reason about interventions, improve from experience, and remained aligned and monitorable as they work with physicians. Current directions include sample efficient clinical LLMs, integration of real-world and experimental evidence for causal effect estimation, and physician interaction with AI recommendations.

Education & Training

- 2025–Present **Residency, Internal Medicine**, *Brigham and Women's Hospital*, Longwood, MA
- 2017–2025 **M.D. with Honors in a Special Field, Health, Sciences, and Technology (HST)**, *Harvard Medical School*, Longwood, MA,
Thesis: *Uncovering Bias Mechanisms in Observational Studies*
(Published in *ICML 2026*; see [3].)
- 2019–2023 **Ph.D., Electrical Engineering and Computer Science**, *MIT*, Cambridge, MA,
Thesis: *Towards Precision Oncology: A Predictive and Causal Lens*
MD-PhD Program at Harvard and MIT
- 2016–2017 **M.S., Computer Science**, *Stanford University*, Stanford, CA
- 2012–2016 **B.S., Computer Science**, *Stanford University*, Stanford, CA

Experience

Research

- 2019–2023 **Ph.D. Student**, *Clinical Machine Learning Group*, MIT
Thesis: *"Towards Precision Oncology – A Predictive and Causal Lens"*
Advisor: Prof. David Sontag.
- Developed methods/algorithms to train state space models, transformer-based models, and LLMs for clinical prediction, causal inference, and physician-AI interaction
 - See my [personal page](#) for an updated summary of my research.
- 2016–2017 **M.S., Research Student**, *QIAI Lab & Hazy Research Lab*, Stanford
Advisors: Dr. Daniel Rubin, Prof. Chris Ré
- Worked on methods to automate domain-specific data augmentation for deep learning models
 - Developed machine learning models to improve decision-making and diagnosis of breast cancer from mammography
- 2016–2017 **B.S., Research Intern**, *DIR Group*, Dayton Interventional Radiology
Advisor: Dr. Mubin Syed
- Developed machine learning model trained on temporal blood clot signals to predict tPA effectiveness at clot dissolution for a particular patient
 - Contributed to development of a novel bariatric embolization technique

Industry

2020–2021 **CTO & Co-Founder**, *Humaine*

Conversational AI coaching for training of medical personnel, including medical students, residents, and other trainees.

- Part of MIT Sandbox and Harvard iLab Venture Program

- Top 10 Startup at MIT 100K Accelerate Final, “Best Social Hack” at HackHarvard

2015 **Software Engineer Intern**, *Acupera*, San Francisco

Built a custom programming language and IDE to enable providers to “code” care management plans

Publications

* indicates equal contribution.

Peer-Reviewed Publications

- [1] I. Demirel, L. Shi, Z. Hussain, and D. Sontag, “LLMs can construct powerful representations and streamline sample-efficient supervised learning,” *Advances in Neural Information Processing Systems*, 2026 **(in press)**.
- [2] Z. Hussain, B. D. Lam, F. A. Acosta-Perez, I. B. Riaz, M. Jacobs, A. J. Yee, and D. Sontag, “Evaluating physician-AI interaction for cancer management: Paving the path toward precision oncology,” *ACM Transactions on Computing for Healthcare*, 2026 **(in press)**
Invited for Oral Presentation at ACM SIGCHI Interactive Health Conference 2026 (premier human-computer interaction venue; declined due to clinical service).
- [3] I. Demirel*, Z. Hussain*, P. De Bartolomeis, and D. Sontag, “Uncovering bias mechanisms in observational studies via predictive performance,” in *International Conference on Machine Learning*, pp. 23722–23758, PMLR, 2026.
- [4] Z. Hussain*, E. De Brouwer*, R. Boiarsky, S. Setty, N. Gupta, G. Liu, C. Li, J. Srimani, J. Zhang, R. Labotka, *et al.*, “Joint AI-driven event prediction and longitudinal modeling in newly diagnosed and relapsed multiple myeloma,” *NPJ Digital Medicine*, vol. 7, no. 1, p. 200, 2024.
- [5] I. Demirel, E. De Brouwer, Z. Hussain, M. Oberst, A. A. Philippakis, and D. Sontag, “Benchmarking observational studies with experimental data under right-censoring,” in *International Conference on Artificial Intelligence and Statistics*, pp. 4285–4293, PMLR, 2024.
- [6] A. M. Alaa, Z. Hussain, and D. Sontag, “Conformalized unconditional quantile regression,” in *International Conference on Artificial Intelligence and Statistics*, pp. 10690–10702, PMLR, 2023.
- [7] Z. Hussain*, M.-C. Shih*, M. Oberst, I. Demirel, and D. Sontag, “Falsification of internal and external validity in observational studies via conditional moment restrictions,” in *International Conference on Artificial Intelligence and Statistics*, pp. 5869–5898, PMLR, 2023.
- [8] Z. Hussain*, M. Oberst*, M.-C. Shih*, and D. Sontag, “Falsification before extrapolation in causal effect estimation,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 6161–6174, 2022.
- [9] Z. Hussain*, R. G. Krishnan*, and D. Sontag, “Neural pharmacodynamic state space modeling,” in *International Conference on Machine Learning*, pp. 4500–4510, PMLR, 2021.
- [10] R. K. Karlsson, M. Willbo, Z. Hussain, R. G. Krishnan, D. Sontag, and F. D. Johansson, “Using time-series privileged information for provably efficient learning of prediction models,” in *International Conference on Artificial Intelligence and Statistics*, pp. 5459–5484, PMLR, 2022.

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- [11] M. K. Sana, Z. Hussain, M. H. Maqsood, and P. A. Shah, "Artificial intelligence in celiac disease," *Computers in Biology and Medicine*, p. 103996, 2020.
- [12] A. J. Ratner, H. R. Ehrenberg, Z. Hussain, J. Dunnmon, and C. Ré, "Learning to compose domain-specific transformations for data augmentation," *Advances in Neural Information Processing Systems*, vol. 30, p. 3239, 2017.
- [13] Z. Hussain, F. Gimenez, D. Yi, and D. Rubin, "Differential data augmentation techniques for medical imaging classification tasks," in *AMIA Annual Symposium Proceedings*, vol. 2017, p. 979, American Medical Informatics Association, 2017.
- [14] Z. Hussain, F. Gimenez, D. Yi, and D. Rubin, "Data augmentation for mammography classification," in *Advances in Neural Information Processing Systems ML4H Workshop*, 2016 (**Spotlight**).
- [15] M. I. Syed, R. Gallagher, Z. Hussain, A. Shaikh, P. J. Cain, K. Morar, M. Sebastian, R. Tyrrell, A. Guehl, and R. Erdelyi, "Combination of thrombolysis and glycoprotein iib/iiia inhibition in chronic peripheral thrombosis: A case report," *International Journal of Radiology and Radiation Oncology*, vol. 3, no. 1, pp. 004–006, 2017.
- [16] E. D. SoRelle, O. Liba, Z. Hussain, M. Gambhir, and A. de la Zerda, "Biofunctionalization of large gold nanorods realizes ultrahigh-sensitivity optical imaging agents," *Langmuir*, vol. 31, no. 45, pp. 12339–12347, 2015.

Invited Talks & Presentations

- 2023 **American Society of Hematology (ASH)**, San Diego, CA
ASH Annual Meeting *Abstract*
 Title: Evaluating Physician-AI Interaction for Multiple Myeloma Management: Paving the Path towards Precision Oncology
- 2023 **Stanford University**, Stanford, CA
Stanford MedAI Seminar Series *Talk*
 Title: Benchmarking Causal Effects from Observational Studies using Experimental Data
- 2022 **MIT**, Cambridge, MA
MIT-MGB AI Cures Conference *Poster Presentation*
 Title: Falsification before Extrapolation in Causal Effect Estimation
- 2022 **MIT**, Cambridge, MA
MIT-MGB AI Cures Conference *Poster Presentation*
 Title: Neural Pharmacodynamic State Space Modeling
- 2018 **Dana Farber Cancer Institute**, Boston, MA
Big Data and Precision Medicine Conference *Breakout Session*
 Title: Disease Progression Modeling
Big Data and Precision Medicine Conference (Invited as co-author.)
 Title: Forays into understanding multiple myeloma with machine learning
- 2018 **Harvard Medical School**, Longwood, MA
 Soma Weiss Research Day *Poster Presentation*
 Title: Modeling disease progression in multiple myeloma
- 2017 **AMIA Annual Symposium** *Oral Presentation*
 Title: Differential data augmentation techniques for medical imaging tasks
- 2016 **NeurIPS ML4H Workshop** *Oral Presentation*
 Title: Data augmentation for mammography classification

Teaching & Mentorship

Teaching

2022 Teaching Assistant for 6.871 (Machine Learning for Healthcare) at MIT

Mentorship

- 2019-Present **Resident Tutor**, Winthrop House, Harvard University
Mentored >100 undergraduate pre-medical students on medical school applications, including mock interviews, personal statement review, and individualized advising. Organized community events, both social and academic, for Winthrop students.
- 2021 Mentor, MIT Summer Research Program – Fernando Acosta Perez
Current PhD Student at University of Wisconsin-Madison

Service

Conferences

- 2025 Reviewer, Uncertainty in Artificial Intelligence (UAI)
Top Reviewer Award
- 2022 Reviewer, Neural Information Processing Systems (NeurIPS)
- 2022 Reviewer, International Conference on Machine Learning (ICML)
Top Reviewer Award (Top 10%)
- 2021 Reviewer, International Conference on Machine Learning (ICML)
- 2021 Reviewer, Uncertainty in Artificial Intelligence (UAI)

Journals

- 2021 Reviewer, Journal of Machine Learning Research (JMLR)

Workshops

- 2023 Reviewer, Time Series Representation Learning for Health (ICLR Workshop)

Awards and Honors

- 2025 MD with Honors in a Special Field, *cum laude*
- 2025 Top Reviewer Award (~Top 10%), UAI 2025
- 2022 NIH Ruth L. Kirschstein National Research Service F30 Award Recipient
- 2022 Top Reviewer Award (Top 10%), ICML 2022
- 2019 NDSEG Fellowship Alternate Awardee
- 2016 Award of Excellence
Presented to top 10% of graduating seniors who exemplify leadership in the Stanford community
- 2013 AFRL Repperger Fellow
Competitive grant given to undergraduates to pursue research at the Air Force Research Lab (AFRL).